Nguyen Khac Duy

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SUMMARY

Backend Engineer with 1.5+ years of experience in healthcare IT, backend development, and system integration. Experienced in building RESTful APIs, optimizing PostgreSQL queries, troubleshooting production systems, and designing event-driven workflows. Independently developing a geospatial social platform using NestJS, PostgreSQL/PostGIS, JWT authentication, and Socket.IO.

TECHNICAL SKILLS

Backend: Node.js, NestJS, Express.js, TypeScript, RESTful APIs.
Databases: PostgreSQL, PostGIS, PL/pgSQL, MongoDB, SQL optimization.
Architecture & Security: JWT, Refresh Token Rotation, Authentication Middleware, WebSocket (Socket.IO), Event-Driven Architecture.
DevOps & Tools: Git, Docker, Docker Compose, GitHub Actions (CI), Swagger/OpenAPI.
Frontend: Next.js, React, Mapbox GL.

WORK EXPERIENCE

DH Digital Solutions **01/2025 – 08/2026**

Technical Implementation & Backend Engineer

- Developed Node.js backend modules for bidirectional HIS–EMR synchronization, coordinating API contracts and data transformations with integration partners.
- Optimized PostgreSQL queries and data-fetching functions using composite indexes, EXPLAIN ANALYZE, and PL/pgSQL, reducing query latency by 15%.
- Diagnosed and resolved production database connection pool exhaustion caused by unreleased connections, improving service reliability during peak workloads.
- Developed a .NET-based desktop bridge integrating digital signature hardware with HIS through a local HTTP API and vendor SDK. GitHub
- Built Node.js API services supporting digital-signature integration and testing with Softdreams EasyCA.
- Supported production troubleshooting, bug isolation, log analysis, and acceptance testing of healthcare applications.

PROJECTS

WanderMap - Geospatial Travel & Social Platform **Personal Project**

Tech Stack: NestJS, Node.js, TypeScript, PostgreSQL 16, PostGIS, Next.js 15, Socket.IO, Mapbox, Docker, GitHub Actions. *GitHub*

- **Backend Development:** Designed and developed a modular **NestJS REST API** with **PostgreSQL/PostGIS** persistence, database migrations, request validation, and structured error handling
- **Frontend Development:** Built interactive user interfaces using **Next.js** and **React**, including social feeds, user profiles, and **Mapbox-powered** travel maps, integrating REST APIs and real-time notifications.
- **Geospatial & Social Features:** Implemented location-based data services using **PostGIS** and social features including travel posts, comments, reactions, bookmarks, and follow relationships.
- **Authentication & Security:** Implemented **JWT-based** authentication with rotating refresh tokens, HttpOnly cookies, and authenticated API access.
- **Real-Time Notifications:** Built a **Socket.IO** notification system with JWT-authenticated connections, user-isolated rooms, persisted notification events, unread counters, and reconnection synchronization.
- **Development & CI:** Configured **Docker Compose** for PostgreSQL/PostGIS and **GitHub Actions workflows** for automated code validation, builds, and testing.

HIS-EMR Real-Time Data Synchronization Engine (DH-EMR-Sync)

Tech stack: Node.js, PostgreSQL (PL/pgSQL), RESTful APIs, JWT. *GitHub*

Role: Backend Developer - Integration Engineer

- **Architecture & Event-Driven Sync:** Built a two-way data synchronization engine between HIS and EMR systems using **PostgreSQL LISTEN/NOTIFY** triggers to process real-time change events, replacing traditional polling mechanisms to reduce database overhead and latency.

- **Database Module:** Developed database-side stored procedures **(PL/pgSQL)** in PostgreSQL to standardize schema transformations and return optimized JSON payloads directly, decoupling data formatting from backend application logic and improving query speed.
- **Reliability & Recovery:** Persisted sync notification events to a dedicated outbox table with processed status and retry_count tracking - enabling recovery of undelivered events after service restarts.

EDUCATION

Can Tho University
Computer science - GPA: 3.12/4 *2020 - 2025*

CERTIFICATIONS

- TOEIC 615 (Achieved: February 2023).
- Smart Testing with AI in the CI/CD Pipeline (CUSC).